

**DESIGN AND IMPLEMENTATION OF A REAL-TIME LAN-BASED QUEUING
MANAGEMENT AND STATISTICAL REPORTING SYSTEM FOR THE
BUSINESS PERMIT AND LICENSING OFFICE (BPLO)**

**ROJEAN F. TANDAS
MANRIJAN M. BARBOSA
CARREN NICOLE A. ESTRELLES
JOHN DAVID R. GARCIA
JADE ALLYSA V. RULLODA**

**DON MARIANO MARCOS MEMORIAL STATE UNIVERSITY
SOUTH LA UNION CAMPUS
COLLEGE OF COMPUTER SCIENCE
AGOO, LA UNION**

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Chapter 1

INTRODUCTION

Situation Analysis

Business Permits and Licensing Offices (BPLOs) play an important role in providing government services to individuals and business owners who apply for new business permits, renew existing permits, or process other business-related transactions. The increasing number of clients during business registration and renewal periods requires an organized process for receiving applications, checking requirements, assigning clients to the appropriate service process, and monitoring the progress of transactions. When queuing and client movement are not properly coordinated, clients may experience uncertainty about their position in the queue, while personnel may experience difficulty in monitoring and managing multiple transactions at the same time. The proposed study focuses on the development of an automated queue management system featuring real-time status updates and staff-controlled queue operations for the Business Permits and Licensing Office.

Queue management has been studied as an important component of service systems because the way clients are assigned to available service personnel can directly affect waiting time and service efficiency. Ainbinder, Temnikov, and Allalouf (2024) compared global and local queue structures in systems with workers having different processing rates. Using discrete-event simulation, the study examined how queue architecture and routing policies affect waiting time and workload distribution. The findings emphasized that queue

design and the manner in which tasks are distributed among service personnel can influence the efficiency of a service system. This is relevant to the present study because BPLO operations may involve personnel performing different functions, making appropriate queue assignment and monitoring important. The study can be adopted conceptually by considering how queue information is organized and how clients are directed to available personnel or service windows.

Lee et al. (2022) examined a priority queueing system involving multiple types of customers and different levels of service priority. The study proposed a flexible priority scheme in which service resources could be allocated according to the current queue condition. The researchers emphasized that priority mechanisms can be used to improve resource allocation while maintaining a degree of fairness among different customer groups. This is relevant to the present study because the proposed BPLO system includes a Priority Waiting status for clients who return after being placed in a Return Tomorrow state. The concept of priority queueing can therefore help support the design of a system where authorized personnel can manage exceptional queue conditions without losing control of the overall client flow.

These international studies demonstrate that queue management is not limited to issuing numbers. Queue architecture, resource allocation, service priorities, and the distribution of clients among available personnel can affect waiting time and operational efficiency. The present study adopts these concepts at a simpler and more practical level

by providing BPLO personnel with real-time queue information and staff-controlled functions for calling, completing, skipping, removing, and managing returning clients.

In Asia, queue management has also been applied through web-based and real-time technologies. Chantima and Pochan (2023), from Naresuan University in Thailand, developed a mathematical model for real-time queue management based on equality principles. The researchers compared a conventional first-come-first-served queue with an appointment-based real-time queue and used Microsoft Excel with Visual Basic for Applications (VBA) to analyze waiting times. Their study demonstrated the importance of continuously considering queue conditions when managing client waiting time. The study is useful to the present research because it supports the idea that queue information should be updated according to the current state of the service process rather than relying only on a static queue sequence.

Santoso et al. (2024), from Politeknik Caltex Riau in Indonesia, developed a queue management system for Puskesmas Umban Sari. The system was designed as an Android- and website-based online queueing system to address difficulties in managing patient queues. It included registration and data-recapitulation functions and supported both priority and linear queues. The system was developed using the Rational Unified Process and evaluated through black-box, usability, and performance testing. This study is relevant to the present research because it demonstrates how different queue types can be incorporated into a computerized system and how queue information can be supported by registration and record management. The present study can adopt the concept of priority

and ordinary queues while applying them to BPLO transactions rather than healthcare services.

Fauzi and Suhaimi (2026) developed and evaluated QKlinik, a web-based queue management system for outpatient clinics in Malaysia. The system incorporated online queue-number reservation, QR-code access, real-time waiting-status tracking, and an integrated interface for clinic-related services. The study addressed the need for greater service transparency and improved patient flow through digital queue information. The system is relevant to the present study because it demonstrates the usefulness of web-based real-time queue monitoring and user access to current queue status. These concepts support the proposed BPLO system's use of system-generated printed ticketing and a public display for providing updated queue information.

The Asian studies reviewed indicate that real-time information, priority management, web-based access, and flexible queue structures can improve the organization of service processes. However, these studies were conducted mainly in healthcare and other service environments. The present study applies similar technological concepts to the specific workflow of a local government Business Permits and Licensing Office, particularly for new business permit applications, renewals, and other supported transactions.

The Philippine government has increasingly promoted the digitalization of business permitting and other frontline government services. In 2024, the Anti-Red Tape Authority

(ARTA), together with the Department of the Interior and Local Government, Department of Information and Communications Technology, and Department of Trade and Industry, intensified efforts to support local government units in establishing electronic Business One-Stop Shops (eBOSS). ARTA emphasized that eBOSS is intended to provide more efficient filing of applications and issuance of local business licenses, permits, clearances, and authorizations.

The scale of this national transition is also reflected in reports from the Philippine News Agency. In January 2024, ARTA reported that 630 out of 1,637 LGUs had either fully or partially implemented eBOSS and that assistance would be provided to LGUs to support wider implementation. By February 2024, ARTA was targeting an additional 120 LGUs for full eBOSS compliance. These developments demonstrate the continuing national movement toward digital business permitting and the need for local governments to modernize their service processes.

Dela Merced, Balmeo, and De Castro (2024) examined the client perspective of the Electronic Business One-Stop Shop (eBOSS) among small and medium enterprises in Quezon City. Their study investigated the relationship between the implementation of eBOSS and client satisfaction and identified opportunities for improving the electronic platform. This study is useful to the present research because it demonstrates that digital business permitting should be examined not only from the perspective of government personnel but also from the perspective of clients who use the service. The present study

similarly considers both clients and BPLO personnel as intended users of the proposed queue management system.

Espinosa, Serrano, and Balmes (2024) developed an automated enrollment queueing system in a Philippine university to improve waiting-line management and operational efficiency. The study examined the weaknesses of the existing enrollment process and developed a computerized queueing system intended to reduce overcrowding, minimize waiting-related difficulties, reduce employee workload, and improve service delivery. Although the setting was an educational institution rather than a government office, the study provides a relevant Philippine example of how automation can organize client flow and reduce manual queue management.

Alegado and Himo (2022) developed a Community Request Queue Management System for the Local Government Unit of Cabanatuan City. The study addressed difficulties involving manual storage of information, retrieval of records, and service arrangements within a local government setting. The system followed the stages of requirements analysis, system design, development, testing, deployment, and maintenance. This study is particularly relevant because it demonstrates the application of a computerized queue and records-management approach in a Philippine local government environment. The present study extends this concept by focusing specifically on business permit-related transactions and real-time staff-controlled queue operations.

The national literature shows that Philippine government offices are increasingly moving toward electronic processing and automated service management. However, electronic

business permitting and queue management are not exactly the same function. An eBOSS primarily supports digital business permit processing, while the proposed study focuses on organizing the physical client flow and service sequence within a BPLO. Therefore, the proposed system seeks to complement digital government initiatives by providing a queue management layer that can organize clients, provide real-time queue information, and give staff direct control over queue operations.

At the regional level, a particularly relevant study was conducted by Rodriguez et al. (2025) in Alaminos City, Pangasinan, which is part of Region I. The researchers examined the development and implementation of an advanced queuing system for the Local Government Unit of Alaminos City. The study aimed to streamline government service processes, minimize waiting time, and improve service efficiency and customer experience. Using a mixed-methods approach involving waiting-time analysis, system-performance measures, customer satisfaction surveys, and staff interviews, the researchers reported improvements in waiting time, customer satisfaction, workflow efficiency, and the organization and transparency of the queueing process. The system also allowed personnel to manage transactions and prioritize urgent cases.

The Alaminos City study is particularly significant to the present research because it demonstrates that queue management is already being explored within a local government setting in Region I. Its use of staff interviews and waiting-time analysis also provides a useful approach for the present study's requirements-gathering process. The present research can adopt the concept of combining client feedback and staff requirements

when designing a queue management system. However, the proposed study differs in its specific focus on the Business Permits and Licensing Office of Rosario, La Union and its inclusion of transaction-based tickets, real-time queue status, return-tomorrow handling, and staff-controlled queue operations.

Regional government developments also show continuing digitalization of business permit services. In 2024, the Anti-Red Tape Authority recognized Bacnotan, La Union for fully complying with the eBOSS system. The municipality was reported as the second LGU in the Ilocos Region, after the City of San Fernando, La Union, to pass ARTA's assessment of fully streamlined and digitalized business permit application-to-issuance processes. The eBOSS implementation included a unified application form, electronic tax bills or orders of payment, online uploading of requirements, electronic permits, integrated online payment, and courier integration.

These developments indicate that digital business permit processing and queue-related service improvements are already emerging within Region I. Nevertheless, the existence of digital permit systems does not mean that every municipality uses the same workflow or queue management mechanism. The present study therefore focuses on the actual operational requirements of the selected BPLO in Rosario, La Union rather than assuming that a system used by another LGU can be directly applied to the selected client. Within La Union, local government units have implemented different approaches to business permit processing. In Agoo, La Union, the Municipal Government organized its 2025 Business One-Stop Shop through the Municipal Treasury Office and Business Permit

and Licensing Office. The BOSS brought representatives from different LGU offices and agencies, including the Bureau of Fire Protection, Sangguniang Bayan Office, Municipal Engineering Office, and Agoo Public Safety and Order Office, into one location to facilitate the processing of business permit applications and renewals. The arrangement demonstrates the importance of coordinating several offices and service functions during business permit processing.

The Municipality of Sudipen, La Union also published its 2025 Business One-Stop Shop procedures. The municipality distinguishes requirements for new business applications from renewal transactions. For new businesses, requirements include proof of registration, documents for tax computation, and other applicable permits and clearances. Renewal transactions have different requirements, including the basis for computing business taxes and other supporting documents. The published process instructs clients to submit documents and application forms and wait for the corresponding assessment before payment. This distinction between new and renewal transactions supports the present study's observation that transaction types should be identified at the beginning of the queue process because they may require different requirements and processing procedures.

The Provincial Government of La Union likewise publishes separate requirements and procedures for new and renewal business registration. Its service procedure identifies the submission and review of forms and documents, verification, assessment of current amounts or previous balances, and forwarding of assessment information for payment. This

demonstrates that business-related government transactions involve several stages and require proper coordination between clients and personnel.

These local examples demonstrate that business permit processing in La Union involves different transaction types, document requirements, assessment procedures, and coordination among personnel and government offices. However, the actual process used by the selected BPLO in Rosario, La Union may differ from the procedures implemented by other municipalities. Therefore, the present study uses the selected BPLO's actual workflow as the primary basis for system requirements.

Based on the information gathered for the present study, clients of the selected BPLO may initially approach personnel for inquiry, submission of requirements, or assistance in obtaining a queue ticket. The first service window plays an important receiving role because personnel assigned to the window accommodate clients and receive applications. The queue is currently monitored and managed by employees through computer-based information, while clients can monitor their queue position through the available display. When a client's number is called, the employee provides the instruction for the client to proceed to the appropriate service point.

The existing situation also indicates that new and renewal transactions may currently pass through similar stages even though their requirements can differ. A new business permit application requires the submission of the required documents and completion of applicable forms. In contrast, a renewal transaction may follow a simpler

procedure when there are no changes in important business details such as ownership, line of business, business name, or business address. In such cases, existing information may be used and the transaction may proceed toward assessment after the required forms and approvals have been completed. This difference is important in designing the proposed queue management system because the selected transaction type should help determine the appropriate queue and service process.

The BPLO also requires clients processing regular permit transactions to accomplish a security checklist that serves as an indicator related to competitiveness. Consequently, the queue system should be designed around the actual workflow and requirements of the office instead of treating all clients as having exactly the same service process.

The literature and local context reveal several important points. First, queue management systems can improve the organization of clients and service resources. Second, real-time queue information can improve transparency and help clients determine when they should proceed for service. Third, priority mechanisms can allow personnel to handle exceptional queue conditions. Fourth, Philippine and regional government offices are increasingly adopting digital business permitting and queueing technologies. However, the existing literature and implementations reviewed do not establish a single queue management system that exactly matches the operational requirements of the selected BPLO in Rosario, La Union.

Thus, the gap addressed by the present study is not simply the absence of a digital queueing system. Rather, the study seeks to develop a queue management system that is specifically adapted to the actual BPLO workflow, including transaction-based queue identification, real-time queue monitoring, staff-controlled queue operations, handling of absent clients, return-tomorrow transactions, priority check-in of returning clients, and queue history and statistics. The system is therefore intended to provide a practical solution that connects the client-facing queue process with the actual operational needs of BPLO personnel.

Statement of Objectives

This study aims to design and implement a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permit and Licensing Office (BPLO) to improve client flow, organize queue operations, and support efficient service delivery through data-driven insights.

Specifically, the study seeks to achieve the following objectives:

1. To design and develop an automated LAN-based queue management system that includes:
 - 1.1. A staff-facing ticket generator that prints a unique queue number based on the client's selected transaction type.
 - 1.2. A staff-controlled dashboard that allows authorized personnel to call the next client, complete a transaction, skip a ticket, and manage priority/returning clients.
 - 1.3. A real-time public monitor display that shows the current queue number and the next client to be served.
 - 1.4. A statistical reporting module for administrators to track office trends and employee performance metrics.
2. To test and improve the system's performance, real-time data synchronization, and overall functionality through iterative alpha and beta testing in a local area network environment.
3. To evaluate the acceptability of the developed system in terms of functional suitability, performance efficiency, usability, and reliability based on the ISO 25010 Software Quality Standards, as assessed by the BPLO staff and clients.

Time and Place of Study

This study was conducted during the First Semester of Academic Year 2026–2027. The system design and development took place at Don Mariano Marcos Memorial State University – South La Union Campus, College of Computer Science, in Consolacion, Agoo, La Union, while the system implementation and data gathering were conducted at the Business Permit and Licensing Office (BPLO) of the Municipality of Rosario, La Union.

Definition of Terms

For a better understanding of this research report, the following terms are operationally defined:

Business Permit and Licensing Office (BPLO) refer to the local government office where business registration, permit renewal, and other licensing-related transactions are processed. It is the primary beneficiary and implementation site of the proposed queue management system.

Client refers to an individual or business representative who visits the Business Permits and Licensing Office to inquire, submit requirements, process a new business permit, renew an existing permit, or perform another supported business-related transaction.

Monitor Display refers to the public-facing screen used to display the currently served queue ticket and succeeding queue information so that waiting clients can determine when they should proceed for service

Current Queue Number refers to the queue number presently being served by the assigned service window. It is displayed in real time to inform waiting clients.

Queue History refers to the collection of recorded queue transactions from the system maintained by authorized personnel for monitoring, statistics, reporting, and future reference.

Queue Management System refers to the automated system developed in this study that manages the issuance of queue numbers, monitors queue status in real time, and supports staff-controlled queue operations to improve service efficiency in the Business Permit and Licensing Office.

Queue Number refers to the unique number automatically generated by the system and assigned to a client according to the selected transaction type. It determines the client's order of service.

Queue Status refers to the current condition of a queue ticket within the proposed system. The statuses include Waiting, Serving, Done, Return Tomorrow, and Priority Waiting.

Waiting refers to the queue status assigned to a newly generated ticket that has not yet been called by authorized personnel.

Serving refers to the queue status assigned to a ticket when the client is currently being accommodated or processed by the assigned personnel.

Done refers to the queue status assigned to a ticket when the client's transaction has been completed by the assigned personnel.

Real-Time Status Updates refer to the live display of queue information, including the current queue number, the next queue number, and changes in queue status as transactions are processed.

Service Window refers to the designated counter where authorized personnel process clients' business permit and licensing transactions after their queue numbers are called.

Staff-Controlled Queue Operations refer to the functions available to authorized personnel for managing the queue. These include calling the next queue number, skipping, removing, and completing queue transactions to maintain an organized client flow.

Transaction Type refers to the category of service selected by the client, such as new business registration, business permit renewal, or other licensing-related transactions. It determines the appropriate queue assignment and service process.

Usability refers to the extent to which the queue management system can be effectively, efficiently, and satisfactorily used by its intended users. In this study, usability forms part of the evaluation based on the ISO/IEC 25010 Software Quality Model.

Chapter 2

METHODOLOGY

Research Design

The researchers will utilize a combination of Developmental Research and Descriptive Research designs. Developmental research will be employed to systematically design, develop, and evaluate the Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permit and Licensing Office (BPLO). This approach is appropriate as the primary goal of the study is to create a functional technological solution to an existing problem.

Simultaneously, descriptive research will be used during the evaluation phase to assess the system's acceptability, efficiency, and usability. This will involve surveying the end-users (BPLO staff and citizens) to describe their experience and gather quantitative feedback on the system's performance compared to the traditional manual queueing method.

Materials and Procedures

The development of the system will utilize the following hardware and software tools:

- **Hardware:** Laptops/Desktops for development and staff usage, a standard printer for ticket generation, a Smart TV or monitor for the queue display, and local area network (LAN) routers.
- **Software (Tech Stack):**

- Frontend: React.js, HTML5, Vanilla CSS for the user interfaces (Staff Dashboard, Admin Panel, TV Display).
- Backend: Node.js with Express.js framework for handling API requests.
- Database: Prisma ORM with a relational database (SQLite/PostgreSQL) to store ticket records, user accounts, and analytics.
- Real-time Communication: Socket.io for live updates between the Staff Dashboard and the TV Display.
- Development Tools: Visual Studio Code (VS Code), Git for version control, and web browsers (Chrome/Edge) for testing.

Software Development Life Cycle (SDLC)

The study will adopt the Agile Methodology as its SDLC model. Agile allows for continuous iteration, testing, and feedback throughout the development process.

1. **Requirements Gathering:** Interviewing BPLO staff to understand the current bottleneck and defining the system features (e.g., System-generated printed ticketing, Staff Dashboard, Return Tomorrow priority feature).
2. **Design:** Creating wireframes, Entity-Relationship Models, Data Flow Diagrams, and Use-Case diagrams.
3. **Development:** Writing the code for the frontend interfaces and backend APIs.
4. **Testing:** Conducting unit testing, integration testing, and simulated user testing to ensure real-time socket connections work properly.
5. **Deployment & Review:** Deploying the system on the local BPLO network and gathering user feedback for further refinements.

Survey Questionnaire (ISO 25010 Based)

To evaluate the system, a customized survey questionnaire based on the ISO 25010 Software Quality Standards will be distributed. The questionnaire will cover:

1. **Functional Suitability:** Does the system generate tickets and call numbers accurately?
2. **Performance Efficiency:** Is the system fast and responsive?
3. **Usability:** Is the printed ticket format clear and is the TV Display easy for citizens to follow? Is the dashboard intuitive for staff?
4. **Reliability:** Does the system maintain data integrity without crashing?

Answering the Statement of Objectives

- **Objective 1 (Design and Develop):** Addressed through the Agile development process, utilizing the React.js and Node.js tech stack.
- **Objective 2 (Test and Improve):** Addressed by conducting alpha and beta testing with the system, utilizing real-time socket events and fixing identified bugs.
- **Objective 3 (Evaluate Acceptability):** Addressed by distributing the survey questionnaire to the respondents and analyzing their feedback.

Data Gathering

To determine the system's effectiveness and acceptability, data will be gathered after the system is deployed for a trial period. The researchers will distribute the survey questionnaires to two groups: the BPLO staff who operate the system and print tickets, and the citizens/clients who receive their printed tickets and monitor the TV display.

Because the citizen population varies daily, a convenience sampling method will be used for the clients, while total enumeration will be used for the BPLO staff.

Table 1. Distribution of Respondents

Respondents	Total Population N	Sample Population n
BPLO Staff	10	10
Citizens / Clients	150	50
TOTAL	160	60

Data Analysis

To analyze the data gathered from the survey questionnaires, the researchers will use Weighted Mean. This statistical tool is appropriate for determining the average response of the participants based on a Likert Scale.

The respondents will rate the system's usability, efficiency, and reliability using a 5-point Likert Scale:

5 - Strongly Agree (Excellent)

4 - Agree (Very Good)

3 - Neutral (Good)

2 - Disagree (Fair)

1 - Strongly Disagree (Poor)

The formula for the weighted mean will be used to interpret the overall acceptability of the queuing management and statistical reporting system.

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